

Integration by partial fractions and by parts

Cambridge insert: two techniques that turn an impossible integral into two easy ones.

LESSON 74–76

3 × 40 MIN

ALGEBRA 11, §3.2 (EXTENSION)

P2 · 8.1–8.3

LESSON CLOCK

20'	30'	30'	30'	26'	4
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Why split a fraction	20 min
Partial fractions	30 min
Integrating them	30 min
By parts	30 min
Practice	26 min
Homework	4 min

LEARNING OBJECTIVES

- Split a proper rational function into partial fractions.
- Integrate using partial fractions.
- State and apply the integration-by-parts formula.
- Choose which factor to differentiate and which to integrate.

TEXTBOOK

National	pp. 357–364
Cambridge	Pure Mathematics 2 & 3, pp. 160–178
Workbook	P2 Exercise 8A–8C

01 Explanation

Why split a fraction

$\int \frac{5x-1}{x^2-1} dx$ has no obvious antiderivative. But the integrand can be written as a sum of two much simpler ones:

$$\frac{5x-1}{x^2-1} = \frac{2}{x-1} + \frac{3}{x+1}$$

and each of those integrates to a logarithm at sight.

THE ONE STANDARD INTEGRAL YOU NEED

$$\int \frac{1}{ax+b} dx = \frac{1}{a} \ln|ax+b| + C$$

Every partial-fraction integral reduces to copies of this.

Splitting the fraction

The fraction must be **proper** — numerator of lower degree than the denominator. If it is not, divide first.

DENOMINATOR	FORM OF THE SPLIT
$(x - a)(x - b)$	$\frac{A}{x - a} + \frac{B}{x - b}$
$(x - a)^2$	$\frac{A}{x - a} + \frac{B}{(x - a)^2}$
$(x - a)(x^2 + c)$	$\frac{A}{x - a} + \frac{Bx + C}{x^2 + c}$

The cover-up method. To find A in $\frac{5x - 1}{(x-1)(x+1)}$, cover the factor $(x - 1)$ and put $x = 1$ into what remains: $\frac{5(1) - 1}{1 + 1} = 2$. Similarly $B = \frac{5(-1) - 1}{-1 - 1} = 3$. Two substitutions, no simultaneous equations.

Integrating them

$$\int \frac{5x - 1}{x^2 - 1} dx = \int \left(\frac{2}{x-1} + \frac{3}{x+1} \right) dx$$

$$= 2 \ln|x - 1| + 3 \ln|x + 1| + C$$

THE MODULUS SIGNS MATTER

$\ln$ is defined only for positive arguments; $\ln|x - 1|$ covers both sides of $x = 1$. Dropping the bars loses a mark and, in a definite integral, can lose the answer.

A repeated factor gives a term that is **not** a logarithm: $\int \frac{1}{(x-2)^2} dx = -\frac{1}{x - 2} + C$

— the power rule, not the logarithm.

Integration by parts

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

It is the product rule integrated. Its purpose is to **replace one integral by an easier one** — so the choice of u decides whether it helps.

CHOOSING u

Take u to be the factor that gets **simpler when differentiated**: a power of x before an exponential; a logarithm before anything.

Example. $\int x e^x dx$: take $u = x$, $\frac{dv}{dx} = e^x$. Then $\int x e^x dx = x e^x - \int e^x dx = e^x(x - 1) + C$.

Example. $\int \ln x dx$: take $u = \ln x$, $\frac{dv}{dx} = 1$. Then $\int \ln x dx = x \ln x - \int 1 dx = x \ln x - x + C$ — an integral with only one factor, done by parts.

A BAD CHOICE MAKES IT WORSE

With $u = e^x$ in the first example, the new integral has x^2 in it. If the second integral is harder than the first, swap the choice and start again.

02 Worked examples

EXAMPLE 1 Split $\frac{5x - 1}{x^2 - 1}$ into partial fractions.

① $x^2 - 1 = (x-1)(x+1)$

② Cover $(x-1)$, put $x = 1$: $\frac{4}{2} = 2$.

③ Cover $(x+1)$, put $x = -1$: $\frac{-6}{-2} = 3$.

ANSWER

$$\frac{2}{x-1} + \frac{3}{x+1}$$

EXAMPLE 2 Find $\int \frac{5x - 1}{x^2 - 1} dx$.

① Use the split above.

② $\int \frac{2}{x-1} dx = 2 \ln|x - 1|$

③ $\int \frac{3}{x+1} dx = 3 \ln|x + 1|$

ANSWER

$$2 \ln|x - 1| + 3 \ln|x + 1| + C$$

EXAMPLE 3 Find $\int x \cos x dx$.

① $u = x, \frac{dv}{dx} = \cos x.$

x simplifies when differentiated.

② $v = \sin x, \frac{du}{dx} = 1.$

③ $x \sin x - \int \sin x dx$

④ $= x \sin x + \cos x + C$

ANSWER

$$x \sin x + \cos x + C$$

03 Interactive model

Do the cover-up method on the board twice — it is faster than any alternative.

This model could not start: Cannot read properties of undefined (reading 'start')

04

Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Partial fractions	Elementar kasrlar	Простейшие дроби
Proper fraction	To‘g‘ri kasr	Правильная дробь
Improper fraction	Noto‘g‘ri kasr	Неправильная дробь
Linear factor	Chiziqli ko‘paytuvchi	Линейный множитель
Repeated factor	Takrorlanuvchi ko‘paytuvchi	Кратный множитель
Integration by parts	Bo‘laklab integrallash	Интегрирование по частям
Cover-up method	Yopish usuli	Метод вычёркивания
Reduction	Kamaytirish	Понижение
Natural logarithm	Natural logarifm	Натуральный логарифм

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05

Quick check
Check yourself

$\int \frac{1}{x-3} dx$ is:

$$\ln(x-3) + C$$

$$\ln|x-3| + C$$

$$-\frac{1}{(x-3)^2} + C$$

$$\frac{1}{2}(x-3)^2 + C$$

$\frac{1}{(x-1)(x+2)}$ splits into:

$$\frac{A}{x-1} + \frac{B}{x+2}$$

$$\frac{Ax+B}{(x-1)(x+2)}$$

$$\frac{A}{(x-1)(x+2)}$$

it does not split

Integration by parts is:

the chain rule reversed

the product rule reversed

the quotient rule reversed

a new rule

For $\int x e^x dx$ take u to be:

$$e^x$$

$$x$$

$$x e^x$$

either

$\int \ln x dx$ equals:

$$\frac{1}{x} + C$$

$$x \ln x - x + C$$

$$\ln x + C$$

$$\frac{(\ln x)^2}{2} + C$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $\int \frac{1}{x-2} dx$

ANSWER $\ln|x-2| + C$

2. $\int \frac{1}{2x+1} dx$

ANSWER $\frac{1}{2} \ln|2x+1| + C$

3. $\int \frac{3}{x} dx$

ANSWER $3 \ln|x| + C$

4. $\int \frac{1}{(x-1)^2} dx$

ANSWER $-\frac{1}{x-1} + C$

5. Split $\frac{1}{x(x+1)}$

ANSWER $\frac{1}{x} - \frac{1}{x+1}$

6. $\int x e^x dx$

ANSWER $e^x(x-1) + C$

7. $\int \ln x dx$

ANSWER $x \ln x - x + C$

Medium · the standard question

7 PROBLEMS

1. Split $\frac{5x-1}{x^2-1}$

ANSWER $\frac{2}{x-1} + \frac{3}{x+1}$

2. $\int \frac{5x-1}{x^2-1} dx$

ANSWER $2 \ln|x-1| + 3 \ln|x+1| + C$

3. Split $\frac{3x+5}{(x+1)(x+3)}$

ANSWER $\frac{1}{x+1} + \frac{2}{x+3}$

4. $\int x \cos x dx$

ANSWER $x \sin x + \cos x + C$

5. $\int x \sin x dx$

ANSWER $-x \cos x + \sin x + C$

6. $\int x^2 e^x dx$

ANSWER $e^x(x^2 - 2x + 2) + C$

7. Split $\frac{4}{x^2-4}$

ANSWER $\frac{1}{x-2} - \frac{1}{x+2}$

Hard · stretch and reason

7 PROBLEMS

1. Split $\frac{x+7}{(x-1)(x+2)^2}$

ANSWER $\frac{\frac{8}{9}}{x-1} - \frac{\frac{8}{9}}{x+2} - \frac{\frac{5}{3}}{(x+2)^2}$

2. $\int \frac{x^2+1}{x^2-1} dx$

ANSWER $x + \ln|x-1| - \ln|x+1| + C$

3. $\int x \ln x dx$

ANSWER $\frac{x^2}{2} \ln x - \frac{x^2}{4} + C$

4. $\int x^2 \ln x dx$

ANSWER $\frac{x^3}{3} \ln x - \frac{x^3}{9} + C$

5. $\int_2^3 \frac{1}{x(x-1)} dx$

ANSWER $\ln \frac{4}{3}$

6. $\int e^x \sin x dx$

ANSWER $\frac{e^x}{2} (\sin x - \cos x) + C$

7. Explain why the cover-up method works

ANSWER Multiplying by the factor and substituting its root kills the other terms

07 Homework

Homework — 6 tasks

Modulus signs inside every logarithm; state which factor you chose as u and why.

1. Split $\frac{7x-1}{x^2-1}$ and $\frac{2x+5}{(x+1)(x+4)}$ into partial fractions.
2. Find $\int \frac{7x-1}{x^2-1} dx$.
3. Find $\int \frac{1}{(x-3)^2} dx$ and explain why it is not a logarithm.
4. Find $\int x e^{(2x)} dx$.
5. Find $\int x^2 \ln x dx$.
6. Explain in three sentences how you decide which factor to call u .